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Robustness of Robotic Manipulation: Foundations and Frontiers

Yifei Dong[1,2], Zhanyi Sun[3], Lujie Yang[4], Manuel Baum[5], Shuran Song[3], Florian T. Pokorny[2], and Xianyi Cheng[1]

Abstract

Humans and animals exhibit remarkable robustness in physical manipulation, yet robots remain far behind. Progress toward human-level manipulation robustness is hindered by the absence of a unified and systematic understanding: different subfields frame robustness in distinct ways, often leaving the concept ambiguous and limiting deeper analysis as well as communication across research areas. This review paper presents a systematic study of manipulation robustness. We begin with a formal definition, characterizing robustness as the degree to which a manipulation system can achieve its goal in the presence of uncertainty and variation. Building on this definition, we introduce general formulations of manipulation robustness from probabilistic or control-theoretic perspectives. We then develop a taxonomy of manipulation robustness mechanisms spanning perception, planning, control, policy learning, and hardware, and illustrate each category through representative seminal works. In addition, we review existing metrics and evaluation methods for quantifying manipulation robustness. We highlight recurring principles of manipulation robustness, strategies for failure management, and broader lessons for designing robust manipulation systems. Finally, we discuss open problems and future directions toward achieving human-level robustness in robotic manipulation.

Keywords

Review, Manipulation Robustness, Manipulation under Uncertainty

1 Introduction

Manipulation in the real world is inherently uncertain (Mason 2018). Object pose may be partially observed, friction and compliance may be poorly modeled, contact events are discontinuous and difficult to predict, and task instances may vary substantially across episodes. Robust manipulation is therefore not simply a matter of executing a nominal plan accurately in an idealized environment. Instead, it is the ability to achieve task goals despite such uncertainty and variation.

Biological systems provide compelling evidence that robust manipulation is possible. Humans handle these uncertainties and variations without deliberate reasoning, drawing on deeply ingrained sensorimotor strategies. For instance, humans regulate grip forces to prevent objects from slipping in the hand. When grasping objects such as fruits (Fig. 2, top-left), humans apply grip forces with a safety margin above the expected slip threshold. Under varying or dynamic conditions, this margin is adjusted accordingly (Hadjiosif and Smith 2015), preventing grasp failure even in uncertain environments.

Robust behaviors are also generally observed in animals. Sea otters, for example, juggle pebbles on their bellies (Foster-Turley and Markowitz 1982) (Fig. 1-a). They maintain this behavior despite turbulent disturbances and across wide variations in object shape, mass, and body posture. Rats adjust the speed and pattern of whisking with their vibrissae to increase information about the shape and texture of contacted objects (Fig. 1-b), demonstrating how exploratory behavior reduces uncertainty about the environment (Grant et al. 2009). Even far less complex organisms exhibit robust interaction with their environment. For example, coordinated ciliary motion in paramecium generates fluid flows that draw and capture food particles (Fenchel 1980) (Fig. 1-c). Without complex visual perception, the organism can handle uncertainty in particle location using only mechanosensory cilia in highly dynamic fluid environments. These examples suggest that robustness in physical interaction is a fundamental property of biological systems (Kitano 2004).

In contrast, today's robotic systems manipulate objects with a level of robustness far below that of humans and many animals (Mason 2018; Billard and Kragic 2019). Even routine behaviors such as pick-and-place, which humans perform effortlessly, remain challenging for robots to execute reliably in the presence of contact uncertainty (Rodriguez 2021). Although impressive demonstrations from academia and industry have shown remarkable capabilities under controlled conditions, many of these systems degrade significantly when deployed outside carefully engineered

[1]Duke University, USA
[2]KTH Royal Institute of Technology, Stockholm, Sweden
[3]Stanford University, USA
[4]Massachusetts Institute of Technology, USA
[5]Organifarms, Germany

Corresponding author:
Yifei Dong, KTH Royal Institute of Technology, Stockholm, Sweden.
Email: yifeid@kth.se

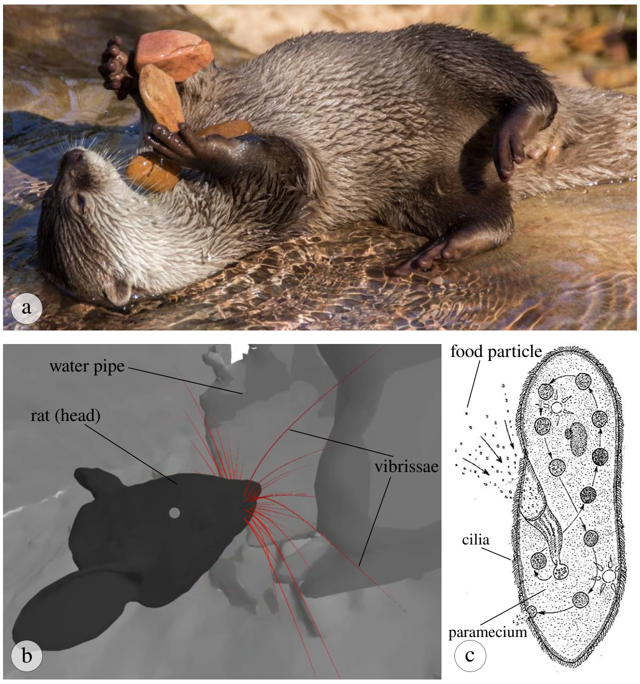

Figure 1. Examples of robust physical interaction observed in biological systems, illustrating the gap between biological and current robotic manipulation robustness. (a) Sea otters juggle stones on their bellies, maintaining control despite variations in rock shape, mass, and environmental disturbances © (Bleckly 2020). (b) Rats actively explore their surroundings by whisking objects with their vibrissae, gathering information to reduce uncertainty about object shape and location (Zweifel et al. 2021) (© PNAS). (c) Paramecium generates ciliary flows that draw and capture food particles despite uncertainty in particle location © (Mackean 2009).

environments. Ultimately, robotic manipulation must operate in open, time-varying environments characterized by significant uncertainty.

This evident gap between robotic and biological manipulation robustness motivates this review. A major obstacle to progress is the absence of a clear definition and overarching framework for manipulation robustness. Each subfield discusses robustness in its own terms, often restricted to a narrow domain and leaving the concept implicit or underspecified. To address this issue, we propose a shared conceptual language and taxonomy to connect existing research threads and enable the accumulation of insights across subfields.

We also identify several foundational but unanswered questions: How is manipulation robustness uniquely positioned within the broader study of robustness in robotics? How can the extensive literature on this topic be organized systematically? How should robustness be quantified and measured? What mechanisms and principles enable robust manipulation? And what explains the extraordinary robustness observed in human and animal manipulation? This review calls attention to these questions and encourages the field to pursue a deeper and more systematic understanding of manipulation robustness.

Manipulation robustness does not seem to arise from a single mechanism but rather from the coordination of mechanisms across perception, planning, control, policy learning, and hardware, calling for joint research across subfields. Recurring principles such as uncertainty reduction, toleration of uncertainty and variation, failure management, etc., appear as common foundations of robustness. We also clarify how robustness is distinct from related concepts such as stability and generalizability, examine whether data alone can deliver robustness, and discuss challenges in measuring and comparing robustness. In the following, we begin by introducing a formal definition and a mathematical formulation of manipulation robustness. We then present a systematic review of existing research across subfields. Next, we analyze current metrics and evaluation protocols, outlining their strengths and limitations. Finally, we provide a guideline for practitioners, highlight open challenges, and suggest future directions toward achieving human-level manipulation robustness.

2 Definition of Manipulation Robustness

This section introduces the concept of robustness progressively. We begin with general definitions of robustness across disciplines, narrow the scope to robotics, and finally formalize the notion of manipulation robustness.

2.1 Robustness Definitions

Despite its frequent use, the notion of robustness is often interpreted differently across contexts and disciplines. Several domain-specific efforts have attempted to formalize robustness in biology (Kitano 2004), machine learning (Braiek and Khomh 2025), and robotics (Baum 2024). We provide representative definitions below for illustration.

Definition 1. Biological robustness. Robustness is a property that allows a system to maintain its functions despite external and internal perturbations (Kitano 2004).

Definition 2. Machine learning robustness. The robustness of machine learning models denotes the capacity of a model to sustain stable predictive performance in the face of variations and changes in the input data (Braiek and Khomh 2025).

Definition 3. Robotic robustness. Robustness specifies the degree to which a behavior can fulfill its task despite being challenged by an adversity (Baum 2024).

Synthesizing these perspectives reveals a common theme: robustness is inherently *context-dependent*. It is not an absolute property of a system, but a relational one. It is defined only with respect to what the system is intended to achieve and the conditions under which it operates. Without context, claims of robustness become ill-posed and difficult to compare across domains or methods.

To make this dependence explicit, we identify four foundational dimensions that together define the context of robustness: *goal*, *challenge*, *mechanism*, and *evaluation*. With these dimensions, we define robustness as follows:

Definition 4. Robustness. Robustness refers to a system's ability to achieve a given *goal* under specified *challenges*, enabled by particular *mechanisms*, and assessed through quantitative *evaluation*.

This structure applies broadly across disciplines. For example, in machine learning, robustness typically refers to maintaining predictive performance (*goal*) under distribution shifts or adversarial perturbations (*challenge*), achieved through techniques such as adversarial training (*mechanism*), and evaluated using quantitative metrics such as worst-case accuracy, or performance degradation under controlled perturbations (*evaluation*).

2.2 Robustness in Robotics

What distinguishes robotics is the embodied physical interactions and tight integration of subsystems. *Goals*: Because robotic systems integrate sensing, mechanics, planning, control, actuation, and learning, robustness must often be addressed both within and across the components. Thus, goals may be defined at different levels of abstraction, ranging from high-level task completion (e.g., object transport or assembly) to component-level objectives such as state estimation accuracy or trajectory tracking performance. *Challenges*: Direct interaction with the physical world exposes robots to uncertainty in perception and contact dynamics, sensing noise and external disturbances, as well as variations in system parameters and environmental conditions. *Mechanisms*: Robustness is realized through complementary passive and active strategies spanning hardware and algorithms. Passive strategies, such as compliant hardware and soft materials, physically tolerate disturbances. Active strategies, such as motion planning, feedback control, and learning methods, proactively mitigate model match or compensate for noises or disturbances. *Evaluation*: Robustness in robotics is commonly evaluated on task execution success rates, complemented by more fine-grained metrics such as stability measure, control convergence, or theoretical guarantees, defined at the level of individual tasks or system components.

2.3 Robotic Manipulation Robustness

Manipulation is characterized as “an agent’s control of its environment through selective contact” (Mason 2018), which makes manipulation robustness a distinct subclass of robotic robustness, with unique challenges arising from contact interactions among the robot, manipulated objects, and the environment.

Manipulation is dominated by multi-body contact interactions, which induce dynamics that are high-dimensional, hybrid, discontinuous, constrained, and therefore difficult to model accurately. These interaction dynamics constitute a major *challenge*, as small variations in geometry, contact conditions, or material properties can lead to qualitatively different outcomes. Perception introduces additional challenges in manipulation because effective contact interaction depends on estimating interaction-relevant states that are largely external to the robot, such as object pose, insertion depth, local contact geometry, material properties, and deformation. Many of these quantities are only partially observable, visually ambiguous, or occluded during contact. Therefore, it has substantially higher demands on perception than tasks such as locomotion, where the robot primarily regulates its own state. Together, these factors complicate the design of robust *mechanisms* in manipulation planning, control, and learning.

Moreover, real-world manipulation encompasses a broad spectrum of task specifications, in which robots must achieve diverse objectives across domestic, industrial, and open-world environments. Unlike locomotion, manipulation relies on multi-body contact interactions among the robot, manipulated objects, and the environment that actively change the state of the world, leading to a combinatorial diversity of manipulation *goals*. Finally, the discontinuous and task-dependent nature of contact interactions makes the definition of general *evaluation* criteria and the establishment of formal robustness guarantees particularly difficult.

We refine Definition 4 for manipulation by specifying the four foundational dimensions as follows:

- *Goals*: Robotic manipulation robustness is defined relative to manipulation goals, which involve achieving desired object states through contact interactions, potentially within task-specific tolerance levels.
- *Challenges*: Challenges arise from uncertainty and variation, which we categorize into epistemic uncertainty, aleatoric uncertainty, and episodic variation. The importance of uncertainty has been recognized since the earliest days of robotics in the 1950s (Goertz 1952), which remains a central theme in robotics research (Mason 2012). Specifically, epistemic uncertainty stems from incomplete or inaccurate knowledge of system properties and can, in principle, be reduced through additional data or modeling. Aleatoric uncertainty reflects irreducible randomness, such as observation noise or stochastic transition disturbances. Episodic variation refers to systematic differences in the physical world that remain fixed within a single task execution but change across task episodes, such as variations in object identity, environment configuration, or robot embodiment (Cui and Trinkle 2021).
- *Mechanisms*: Mechanisms mitigate uncertainty and variation in contact interactions and improve task goal achievement through strategies spanning perception, planning, control, learning, and hardware design.
- *Evaluation*: Evaluation methods define quantitative criteria for measuring manipulation performance and robustness, enabling principled comparison across approaches.

Despite the differences, manipulation robustness shares the same fundamental nature as other robustness problems. This allows transfer of principles from related areas, such as domain randomization or data augmentation from machine learning, and analogies to locomotion robustness (Shi et al. 2024).

2.4 Robustness and Related Concepts

Robustness often intersects with but differs from other fundamental concepts, including safety, stability, generalizability, etc. *Safety* concerns preventing harm to humans, the environment, or the robot itself, and is critical in domains such as autonomous driving, navigation, and human–robot interaction (Brunke et al. 2022). *Stability*, common in control and grasping, refers to maintaining or converging to a desired state under perturbations. Robustness extends beyond these notions, encompassing not just safe or stable operation but the broader pursuit of general task goals under uncertainty or

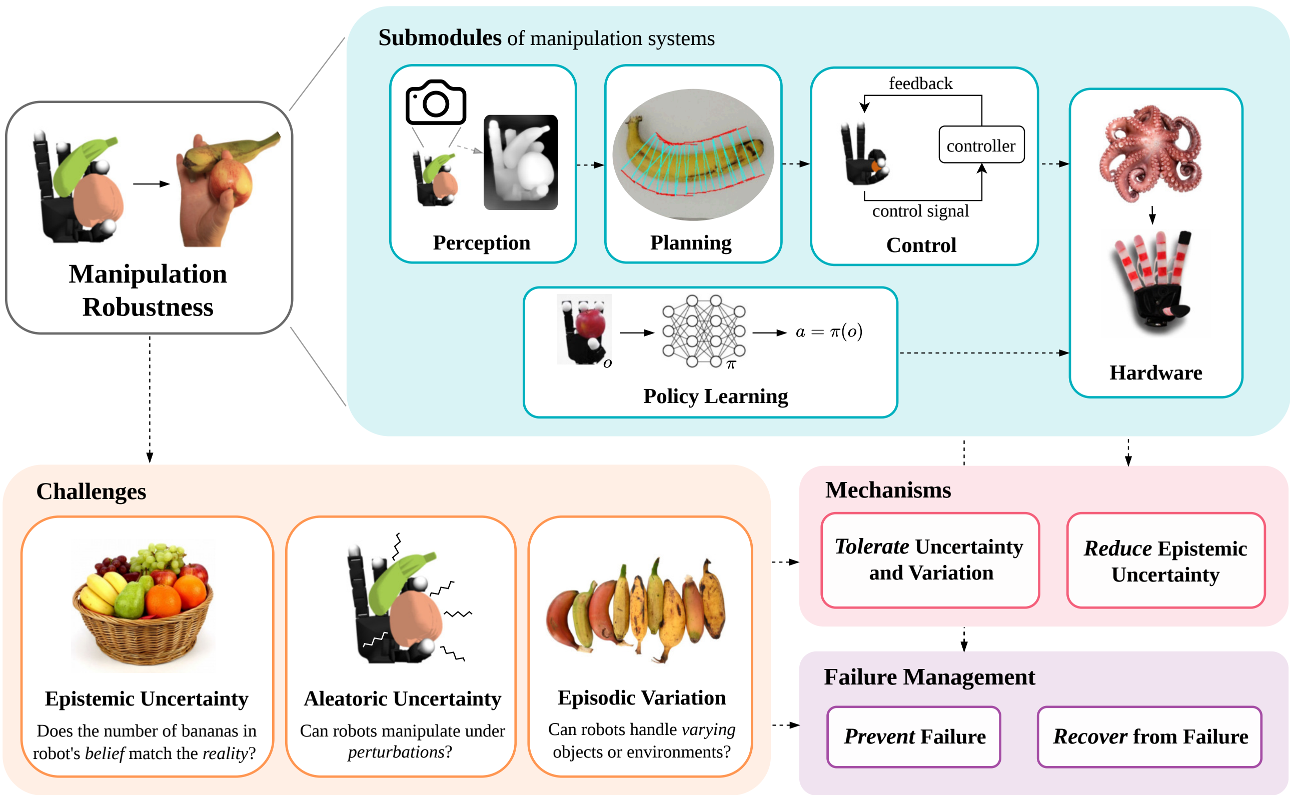

Figure 2. Overview of the survey on robustness of robotic manipulation. We use a fruit manipulation task as an example. A robotic manipulation system has primarily five submodules: perception, planning, control, policy learning, and hardware. Manipulation robustness faces three core challenges: epistemic uncertainty, aleatoric uncertainty, and episodic variation. To tackle the challenges, there are mechanisms across the submodules. These mechanisms either tolerate uncertainty and variation, or reduce epistemic uncertainty. On the other hand, a robust manipulation system achieves its goals by preventing failure or recovering from it. Photo credits: Fruit manipulation with Allegro and human hand (Lu et al. 2025) (© IEEE); octopus-inspired compliant hand (Sankar et al. 2025) (© AAAS); bananas of varying shapes (Fruit Hunters 2026); basket of fruits (Online Delivery 2024); dexterous apple grasping (Berkeley AI Research 2019); banana grasp synthesis (Cao et al. 2022) (© Elsevier). (Permission to adapt some photos used in this paper has been requested from the original authors.)

variation. Robustness and *generalizability* are frequently discussed interchangeably, even though they capture different aspects of system behavior. Generalization refers to a learned model's ability to handle out-of-distribution data or novel domains, whereas robustness emphasizes measurable and reliable performance within the domain despite uncertainty and variation.

3 Mathematical Formulation of Manipulation Robustness

Given the definition in Section 2, we formulate the general problem of manipulation robustness using a unified partially observable stochastic control model. We then instantiate this model through two dominant perspectives in the literature: probabilistic modeling (POMDPs) and constrained optimization (robust control). The unified model serves as an analytical framework to structure the discussion in the remainder of the paper. It is intended as a conceptual tool for analysis rather than a prescriptive model; it does not imply that a robot must explicitly represent these processes to achieve robust manipulation.

3.1 Unified Formulation

We formalize a manipulation episode as a discrete-time control process over a finite horizon T, denoted by the sequence $\{(X_t, U_t, Y_t)\}_{t=0}^{T}$. Here, $X_t \in \mathcal{X}$ is the system state (encompassing both robot and object generalized coordinates), $U_t \in \mathcal{U}$ is the control input, and $Y_t \in \mathcal{Y}$ is the partial observation.

The evolution of this system is driven by underlying deterministic functions subjected to noise:

$$X_{t+1} = f_\theta(X_t, U_t, W_t), \tag{1}$$
$$Y_t = g_\theta(X_t, V_t). \tag{2}$$

In this framework, manipulation robustness is defined by how well a system achieves a target goal region G despite three fundamental challenges:

Aleatoric Uncertainty The variables $W_t \in \mathcal{W}$ and $V_t \in \mathcal{V}$ represent exogenous process and measurement noise. This captures inherent, irreducible stochasticity in the physical world, such as thermal sensor noise, unmodeled micro-impacts, or random slip during contact.

Epistemic Uncertainty The parameter $\theta \in \Theta$ dictates the true physical and perceptual properties of the task, such as object mass, surface friction, and camera calibration. However, the robot rarely has perfect knowledge of θ

or the true state X_t. It operates using internal estimates $\hat{\theta}$ and $\hat{X}_t$. The discrepancy between reality (θ, X_t) and the robot's belief $(\hat{\theta}, \hat{X}_t)$ constitutes epistemic uncertainty. Unlike aleatoric noise, this uncertainty can theoretically be reduced through exploration or learning.

Episodic Variation While epistemic uncertainty concerns what the robot does not know, episodic variation describes the objective physical differences between separate executions of a task. We define an episodic variation ν as a specific draw of the environment parameters, initial state, and goal region:

$$\nu = (\theta, x_0, G) \in N. \tag{3}$$

3.2 Probabilistic Formulation via POMDPs

Learning-based and probabilistic planning methods treat the noise variables (W_t, V_t) and episodic variations (ν) as probability distributions. Consequently, the dynamics are viewed as stochastic transition and observation kernels,

$$X_{t+1} \sim \mathcal{T}_\theta(\cdot \mid X_t, U_t),\ Y_t \sim \mathcal{O}_\theta(\cdot \mid X_t) \tag{4}$$

Robustness in this paradigm is typically evaluated as the expected performance of a policy π across a distribution of episodes ρ_N:

$$\Gamma(\pi) = \mathbb{E}_{\nu \sim \rho_N}\left[J_\nu(\pi)\right], \tag{5}$$

where $J_\nu(\pi)$ is the expected return of the trajectory under the true parameters θ, despite the policy acting on its imperfect belief $\hat{\theta}$.

3.3 Control-Theoretic View via Constrained Optimization

In contrast, robust control frameworks (e.g., H_∞ or Robust MPC) model aleatoric noise w_t, v_t and episodic variations ν as unknown but deterministic variables strictly confined within bounded sets $\mathcal{W}, \mathcal{V}$, and N. Robustness here is framed as a min-max optimization problem. The goal is to synthesize a policy that guarantees constraint satisfaction and minimizes a cost function $c(x, u)$ against the worst-case possible realizations of the uncertainties:

$$\min_{\pi} \max_{\nu \in N, w \in \mathcal{W}, v \in \mathcal{V}} \sum c(x_t, u_t). \tag{6}$$

While the probabilistic view maximizes average-case reliability across a distribution of environments, the control view provides strict performance and safety guarantees, provided the real-world variations never breach the defined bounds $\mathcal{W}$ and $\mathcal{V}$.

4 Existing Studies on Manipulation Robustness

In this section, we survey four decades of research on robustness in robotic manipulation. We focus on the underlying mechanisms that enable robust performance, illustrated through representative and influential works. To structure this discussion, we adopt a goal–challenge–mechanism taxonomy, summarized in Table 1.

We organize robustness mechanisms into five subfields: perception, planning, control, learning, and hardware (Fig. 2). Each encompasses a set of core mechanisms. For every mechanism, we identify key references and representative manipulation tasks that exemplify its role in achieving robustness. Across these subfields, the primary challenges stem from epistemic uncertainty, aleatoric uncertainty, and episodic variation. At a high level, robustness mechanisms address these challenges through two complementary strategies: (i) reducing epistemic uncertainty (Inglis 2000), by improving the robot's knowledge of the environment, task, or interaction dynamics; and (ii) tolerating aleatoric uncertainty and episodic variation, by compensating for randomness or variation that cannot be reduced. Mechanisms further differ in how they manage failure: some aim to prevent failure proactively, while others focus on recovering from failure once it occurs. In the remainder of this section, we detail these mechanisms within each subfield and analyze their roles in achieving robust manipulation.

4.1 Perception

The goal of perception is to gather sufficient information for a robot to carry out its task robustly. Especially in the context of manipulation, perception faces closely connected aspects of epistemic and aleatoric uncertainty, as well as episodic variation.

Regarding episodic variation, we often require perception to identify task-relevant information such as parameters θ or the initial state $x_0 \in \mathcal{X}_0$ that might only be partially known a priori. Episodic variation is often not critical, as long as it is *known*. When episodic variations are *unknown*, however, we often face challenges regarding epistemic and aleatoric uncertainty.

Aleatoric uncertainty challenges perception for manipulation in at least two regards. First, the different sensors required for manipulation each require different measurement models, subject to different kinds of sensor noise. However, we can also leverage this property in *multimodal perception* (Section 4.1.2) and utilize this complementarity by relying on the best combination of sensors in each situation. Second, sensor noise in manipulation is often heteroscedastic, which means that the noise distribution depends on system state and the robot's actions. However, this opens up opportunities for *active and interactive perception* (4.1.1), where robots can perform actions and manipulate the state of the system to facilitate perception. Another challenging aspect of epistemic uncertainty are outlier measurements, which require *robust estimation* techniques (4.1.3).

Epistemic uncertainty challenges perception for manipulation because task-relevant properties and system dynamics are often unknown initially. Important information, such as the number of objects in a scene, their motion constraints and friction properties can often only be identified online, by means of interactive perception. *Bayesian perception* allows robots to maintain an estimate about the remaining degree of aleatoric uncertainty (Section 4.1.3). Ideally, perception should not maintain a belief over variables that are task-relevant and representations should be *invariant* to other features (4.1.4).

4.1.1 Active and Interactive Perception Many challenges to robot perception can be addressed by using robots'

Table 1. Taxonomy of Manipulation Robustness Mechanisms. Abbreviations: EU = epistemic uncertainty, AU = aleatoric uncertainty, EV = episodic variation, PRE = failure prevention, REC = failure recovery.

Subfield	Mechanism	Challenge	Failure Management	Section
Perception	Active and Interactive Perception	EU/AU	PRE	4.1.1
	Multimodality	EU/AU	PRE/REC	4.1.2
	Bayesian Perception and Robust Estimation	EU/AU	PRE	4.1.3
	Invariance in Perceptual Representations	EU/EV	PRE	4.1.4
Planning	Reasoning over Uncertainties	EU/AU	PRE	4.2.1
	Constraint-aware Planning	EU/AU/EV	PRE	4.2.2
	Motion Strategy and Funneling	AU/EV	PRE	4.2.3
	Closure	AU/EV	PRE	4.2.4
	Topological methods	AU/EV	PRE	4.2.5
Control	Reactivity	AU/EV	PRE/REC	4.3.1
	Predictivity	EU/AU	PRE	4.3.2
	Active Compliance	AU/EV	PRE	4.3.3
	Control Invariance	AU/EV	PRE	4.3.4
Learning	Distribution-centric Robustness	AU/EV	PRE	4.4.1
	Architecture- and Representation-centric Robustness	EU/AU/EV	PRE	4.4.2
	Objective-centric Robustness	AU/EV	PRE	4.4.3
	Adaptation-centric Robustness	EU/AU/EV	PRE/REC	4.4.4
Hardware	Passive Compliance	AU/EV	PRE	4.5.1
	Adhesion	AU/EV	PRE	4.5.2
	Postural Synergy	AU/EV	PRE	4.5.3
	Morphology Adaptation	EV	PRE/REC	4.5.4

capability to act in the world (Aloimonos et al. 1988). Occlusions are a major challenge in computer vision, and we can often address them by simply moving a camera to change its viewpoint (Bajcsy 1988), for example, to support grasp execution (Morrison et al. 2019) (Fig. 3-a). Active camera movement can also increase the robustness of structure and depth estimation. This works even in highly challenging circumstances, such as when an object can only be seen through a mesh-like fabric (Battaje and Brock 2022).

Beyond contactless camera movement, robots can also physically interact with their environment for the purpose of perception. This is called Interactive Perception (Bohg et al. 2017; Xiong et al. 2025), which originates from animals that use exploratory behaviors to gather information about the environment (Chappell et al. 2012). Some object properties, such as mass distribution, are not observable without physical interaction. For example, lifting a milk carton reveals its weight distribution, whereas removing a lid to identify the contents of a box. In cluttered environments, it may be necessary to manipulate obstructing objects to expose the target. In particular, scene segmentation can benefit from pushing actions that induce motion in the scene (Xu et al. 2015). Interactive perception can also reveal motion constraints and kinematic structures of articulated objects (Katz and Brock 2008). However, direct interaction can be costly and risky. A common way to mitigate this is through model-based approaches that predict outcomes of exploratory actions. Such models are often the basis of deliberate exploration strategies, such as entropy minimization or cross-entropy maximization (Baum et al. 2017).

4.1.2 Multimodality Different sensor modalities have complementary strengths and weaknesses. Thus, we can achieve robustness by jointly leveraging multiple modalities, such as vision, touch, audio, and proprioception, to leverage their complementary strengths (Fig. 3-b). If one modality fails, others can still acquire partial information and avoid task failure—humans in darkness, for example, rely on touch. In robotics, vision provides global awareness but is prone to occlusion, while touch conveys rich local information such as stiffness, texture, friction, and local geometry. Audition captures transient events, such as the sound of a snap-fit connection (Li et al. 2023). For instance, Izatt et al. (2017) fused RGB-D with tactile feedback for contact-aware pose tracking robust to occlusion. Similarly, proprioception (resistive bend sensors) complements exteroception (fingertip force sensors) in soft hands to improve grasp robustness under uncertainty (Homberg et al. 2019). This can be understood as redundancy across modalities: the different sensing channels compensate for one another. Redundancy also arises within a single modality, as in distributed sensory systems composed of multiple sensing units, such as whiskers in rodents or compound eyes in insects. Analogously, robotic skins and tactile arrays (Hughes et al. 2018) enhance both sensitivity and precision through dense, distributed sensing.

4.1.3 Bayesian Perception and Robust Estimation Even high-quality sensors cannot fully eliminate epistemic and aleatoric uncertainty. The Bayesian approach addresses this limitation by explicitly representing uncertainty as a belief over the relevant latent variables (Fig. 3-c). Such belief representations enable exploration behavior and allow robot policies to perform cautious control by taking into account how reliable information is. To maintain a belief, probabilistic state estimation integrates observations and the effects of actions over time, as in Kalman and Particle Filters (Thrun 2002). Bayesian methods have been widely applied in tasks such as in-hand object localization with tactile inputs (Chebotar et al. 2014), estimating the state of articulated objects (Petrovskaya et al. 2006) or container fill levels (Chitta et al. 2011).

Sensor input might not only be noisy, but can sometimes be corrupted to an extent where it is completely unusable.

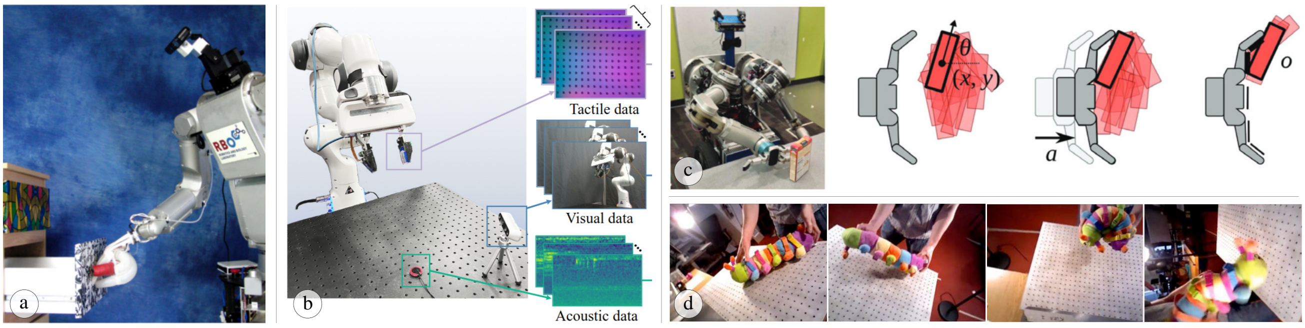

Figure 3. Examples of robustness mechanisms in perception. (a) Interactive perception: the robot interacts with the environment to improve perceptual understanding © (Martín-Martín 2018). (b) Multimodality: perception combining tactile, visual, and acoustic sensing modalities (Li et al. 2023) (© PMLR). (c) Bayesian perception and robust estimation: object pose uncertainty is explicitly represented using Bayesian methods (Koval et al. 2015) (© SAGE). (d) Invariance in perceptual representation: the perception of the toy remains invariant to changes in camera pose, object pose or deformation, and lighting conditions (Florence et al. 2018) (© PMLR).

It is crucial that perception systems for manipulation can identify and ignore such measurements, called outliers. Bayesian approaches are often inherently robust against outliers (O'Hagan 1979), particularly when utilizing heavy-tailed distributions and hierarchical models (West 1984). Beyond Bayesian methods, a range of robust estimation techniques provides built-in mechanisms for outlier rejection (Antonante et al. 2021). Techniques such as RANSAC and M-Estimation have been applied for tasks such as robust 6D object pose tracking from vision (Wen and Bekris 2021) and for visual servoing (Comport et al. 2004).

4.1.4 Invariance in Perceptual Representations Perception for manipulation is challenging because objects need to be reliably detected from different viewing angles and lighting conditions (Fig. 3-d). Before deep learning approaches started to dominate computer vision (Krizhevsky et al. 2012), computer vision pipelines were often based on features such as SIFT (Lowe 2004), SURF (Bay et al. 2006), or ORB (Rublee et al. 2011), which were designed to be invariant to rotation, scale, and possibly other transforms. In deep learning based approaches, such invariance is often achieved using data-augmentation (Krizhevsky et al. 2012) and pooling mechanisms (LeCun et al. 2002). Recent models, such as DINO (Oquab et al. 2023), can learn task-agnostic visual representations that produce features highly robust to clutter, occlusion, and illumination. Language-based scene representations are highly invariant, which has also allowed vision-language-action models to show interesting robustness properties (Zitkovich et al. 2023).

4.2 Planning

Planning operates based on perception and is therefore inevitably affected by its imperfections. In particular, imperfect perception or state estimation induces *epistemic uncertainty*: a mismatch between the true robot–environment parameter θ and the internal model $\hat{\theta}$, and consequently between the true state x_t and the state estimate $\hat{x}_t$. Under such mismatches, trajectories computed by the planner may fail to reach the goal set G. As discussed in Section III, manipulation planning must also contend with aleatoric uncertainty in the dynamics and observation models, as well as episodic variation across task instances.

Existing robust planning mechanisms improve manipulation performance through two complementary strategies. First, it can *reduce* epistemic uncertainty by explicitly reasoning over uncertainty (Section 4.2.1). Second, when uncertainty cannot be substantially reduced, planning can *reduce sensitivity* to disturbances by explicitly searching within safe, constraint-satisfying spaces (4.2.2), by exploiting task mechanics through motion strategies and funnels (4.2.3), or by forming "closures" around target objects (4.2.4).

4.2.1 Reasoning over Uncertainties A direct route to mitigating epistemic uncertainty is to represent it explicitly as a probability distribution over possible outcomes—commonly referred to as a belief over states—and to update this belief online using new observations (Fig. 4-a). In the POMDP formulation, the agent maintains a belief $b_t \in \mathcal{B}(\mathcal{X})$ over the state x_t, conditioned on the history $h_t = (y_{0:t}, u_{0:t-1})$ under the internal model $\hat{\theta}$, i.e., $b_t(x) = \mathbb{P}(x_t = x \mid h_t, \hat{\theta})$. Planning then operates in belief space, with a policy of the form $u_t = \pi(b_t)$, aiming to improve expected task performance across episodic variations ν.

Unlike classical motion planning, belief-space methods explicitly trade off goal progress with information gathering, reducing posterior ambiguity that would otherwise lead to suboptimal actions. For instance, Platt Jr et al. (2010) applied belief dynamics to grasping, synthesizing locally optimal feedback policies in belief space and replanning under unexpected observations. Similarly, Jankowski et al. (2024) incorporated pose and contact-dynamics uncertainty into a belief-space formulation and validated it on planar pushing without sensory feedback. At longer horizons, belief reasoning extends to uncertainty-aware task and motion planning; Kaelbling and Lozano-Pérez (2013) integrated belief propagation with symbolic decision making to address incomplete knowledge of object properties and locations in mobile manipulation.

4.2.2 Constraint-aware Planning In contrast to explicitly reasoning over uncertainties to mitigate them, another approach is to tolerate or reduce sensitivity to them, as they are often irreducible in the form of aleatoric uncertainty or episodic variations. Constraint-aware planning is one such strategy. Specifically, many manipulation goals require changing the object's state through interaction

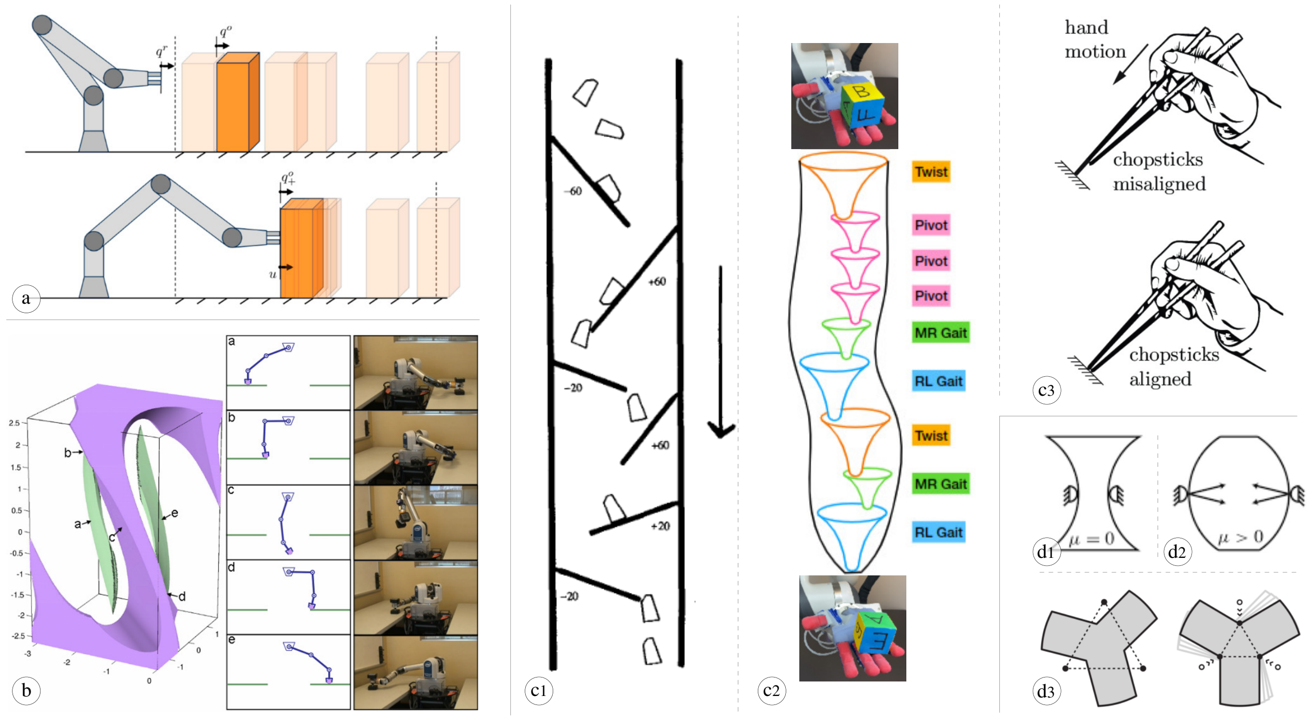

Figure 4. Examples of robust planning mechanisms. (a) Reasoning over uncertainties: planning in belief space (Jankowski et al. 2024), where the belief of the box position is updated after it is pushed by the robot arm (© SAGE). (b) Constraint-aware planning: a collision-free trajectory (a–b–c–d–e) is planned on a constrained manifold defined by kinematic constraints (purple) and the constraint of the dumbbell sliding on the table (green) (Berenson et al. 2009) (© IEEE). (c) Motion strategy and funneling: (c1) funneling on a conveyor belt (Peshkin and Sanderson 2002), where objects with varying poses converge to a single pose through guiding fences (© IEEE); (c2) funneling with in-hand reorientation primitives, where primitives are chained to reach the target cube orientation © (Bhatt et al. 2021); (c3) motion strategy for aligning chopsticks (Shi et al. 2020) (© IEEE). (d) Closure: (d1,d2) force closure and form closure illustrated in a planar two-point-mass-finger scenario (Fox 2020); (d3) caging with three fingers and transition from caging to grasping by contracting the fingers (Rodriguez et al. 2012) (© SAGE).

while satisfying state and input constraints, $x_t \in \mathcal{X}$ and $u_t \in \mathcal{U}$. In configuration space, such requirements induce constraint manifolds (often low-dimensional) that encode collision avoidance, contact constraints, kinematic limits, torque bounds, etc (Fig. 4-b). Epistemic uncertainty in θ and aleatoric uncertainty in (w_t, v_t) can drive the system off these manifolds, causing constraint violation and failure; constrained manifold planning counters this sensitivity by searching directly in safe, constraint-satisfying subsets so feasibility is preserved by construction.

The main challenge in constraint-aware planning is that these manifolds are frequently nonlinear, disconnected, or occupy only an infinitesimal portion of the higher-dimensional configuration space, making them difficult to explore and plan. To address this, Berenson et al. (2009) utilize sampling-based planning directly on constraint manifolds. Siméon et al. (2004) decomposed the configuration space into mode-specific manifolds for transit (non-contact) and transfer (contact) motions, building separate probabilistic roadmaps and linking them through a manipulation graph. Saha and Isto (2007) demonstrated knot-tying with deformable linear objects by planning in the topological space of rope crossings, where constraints encoded at the topological level guaranteed validity under variations in rope geometry and mechanics.

4.2.3 Motion Strategy and Funneling Besides constraint-aware planning, another way to reduce sensitivity to uncertainty is to exploit task mechanics. In *motion strategies* (also termed sensorless manipulation), contact dynamics and gravity provide passive negative feedback that drives objects toward desired configurations, often without exteroceptive sensing (Erdmann and Mason 2002). This contrasts with sensor strategies, which rely on rich observations but are susceptible to perceptual errors. Motion strategies are effective in diverse settings, including tray-tilting to orient objects of unknown pose using only gravity and contact (Erdmann and Mason 2002), push–grasp behaviors in clutter that funnel the target into the gripper while displacing distractors (Dogar et al. 2012), and industrial part feeders whose fences enforce consistent orientation under pose and contact variability (Peshkin and Sanderson 2002) (Fig. 4-c1). More generally, environmental contact and gravity as task mechanics can reduce uncertainties, or *collapse* episodic variations into a predictable, smaller set of outcomes (often termed *extrinsic dexterity* (Dafle et al. 2014)). For example, pressing a loose pair of chopsticks against a tabletop passively aligns their tips through the collision (Fig. 4-c3).

A representative abstraction of motion strategy is *funneling* (Mason 1985). A funnel is a region of attraction $\mathcal{F} \subset \mathcal{X}$ such that, for $x_0 \in \mathcal{F}$ and bounded disturbances $w_t \in \mathcal{W}, v_t \in \mathcal{V}$, the resulting execution converges to the goal set G (i.e., $x_T \in G$), despite pose or geometry uncertainty and model mismatch. Such strategies exploit repeated contacts and environmental constraints to steer outcomes without precise sensing, as in vibrating bowl feeders used for industrial part feeding. The funneling concept traces to pre-image backchaining (Lozano-Perez et al. 1984), which recursively constructs sets of states from which motions succeed under bounded execution errors. It was later extended by LQR-Tree methods (Tedrake 2009), where verified funnels cover the reachable state space to enable robust trajectory planning under model uncertainty. More recently, composing simple in-hand manipulation funnels has shown surprising robustness to variations in object mass and pose (Bhatt et al. 2021) (Fig. 4-c2).

4.2.4 Closure In contrast to motion strategies that steer objects toward desired states, some tasks instead require the object state to be constrained. In such cases, robustness can be achieved through *closure*, which imposes geometric or force constraints on object motion. In many grasping and fixturing tasks, the goal is to maintain the object within an in-hand stable set. Therefore, closure can be viewed as shaping the robot-object interactions such that the object's reachable configurations remain within a bounded subset $G \subset \mathcal{X}$ despite bounded disturbances $w_t \in \mathcal{W}$ in the dynamics $x_{t+1} = f_\theta(x_t, u_t, w_t)$. Force closure captures equilibrium grasps that can resist arbitrary external wrenches via internal contact forces (Howard and Kumar 1996) (Fig. 4-d2). Form closure is a geometric condition in which contact constraints eliminate all object motions even without friction (Bicchi 1995) (Fig. 4-d1). Object closure (caging) further relaxes contact requirements by trapping the object within a bounded region of its configuration space without precise force regulation (Pereira et al. 2004) (Fig. 4-d3). Across these closure methods, the shared principle is to restrict the configuration space so that the object cannot move freely or escape, making the manipulation outcome less sensitive to perception error, control imprecision, and external disturbances.

4.2.5 Topological Methods In contrast to *local* reasoning methods (e.g., force and form closure) in robotics based on an infinitesimal analysis, *topological methods* focus on topological features that are robust under continuous deformation. Motion planning and reasoning methods that consider planning up to a *homotopy class*, provide an example of this approach (Canny 1988; Farber 2003). More recently, *Topological Data Analysis* (TDA) (Zomorodian 2012; Edelsbrunner et al. 2008), provides a bridge between classical topological approaches and the more modern data-driven paradigm. TDA has evolved as a sub-discipline of Machine Learning and features a rigorous methodology to infer global topological information from noisy sampled data by means of the computation of filtrations of simplicial complexes and persistent homology or cohomology groups. This approach has been applied for robust sampling-based reconstruction of configuration spaces and motion planning up to homology groups (Pokorny and Kragic 2015; Bhattacharya et al. 2015) as well as motion clustering (Pokorny et al. 2016) for robotics, for instance. In the context of manipulation, such representations are particularly useful for capturing the global structure of contact modes or configuration spaces, thereby providing robustness to sensing noise and geometric variation.

4.3 Control

A large class of robotic control methods addresses motion control in free space, such as trajectory tracking or point-to-point reaching, where the manipulator operates without contact. Manipulation, however, is fundamentally characterized by the making and breaking of unilateral, frictional contacts, which introduce discontinuities and uncertainties. Manipulation thus requires the regulation of both forces and discrete contact events, posing unique challenges. Control in contact-rich settings must therefore contend not only with external disturbances and actuation errors but also with epistemic uncertainties in contact modeling. Poorly tuned controllers can amplify these difficulties, leading to excessive forces, particularly against rigid environments. Model-based interaction control mitigates such effects by regulating force, torque, or compliance, although these methods often rely on simplified models, demand high computational resources in high-dimensional systems, and remain difficult to generalize. Model-free neural controllers offer a more adaptable alternative, though their performance comes at the expense of data efficiency. Building on these observations, we identify four key mechanisms of robust manipulation control: *reactivity* (Section 4.3.1), *predictivity* (4.3.2), *compliance* (4.3.3), and *invariance* (4.3.4).

4.3.1 Reactivity In manipulation, reactivity denotes the ability to locally respond to unpredictable or unmodeled contact dynamics and environmental variations. Its primary advantage lies in handling unforeseen contact events (Dietrich et al. 2012). Reactive mechanisms are typically grounded in feedback control: in human motor control, for instance, grip force is continuously adjusted through tactile feedback—slip induces higher applied force, while excessive pressure reduces it (Lehman et al. 2013) (Fig. 5-a). Similar principles are applied in robotic manipulation through classical position or force control, such as hybrid position–force schemes for tasks such as peg-in-hole insertion (Beltran-Hernandez et al. 2020), or artificial potential-field methods (Khatib 1986). More recently, neural controllers have provided reactive behavior with fast inference, high adaptability, and generalization. For example, Xue et al. (2025a) demonstrated high-frequency visual–tactile feedback control, which enables robust performance in contact-rich tasks such as peeling a cucumber or wiping a vase under external disturbances. Despite these benefits, reactivity is inherently local and prone to local minima, making it most effective when combined with global planning.

4.3.2 Predictivity Neuroscience suggests that human manipulation relies on the integration of long-term prediction with short-horizon reactivity (Flanagan et al. 2006). Reactivity provides rapid sensory feedback, particularly tactile and visual, to detect mismatches between expected and actual outcomes and correct errors. Predictivity, on the other hand, enables the anticipation of collisions or contact events. Purely local reactive control may fail in contact-rich settings, where abrupt changes in dynamics due to contacts can drive the system into unsafe states. To address this, various control methods incorporate a lookahead structure with reactive schemes. Model-based approaches such as model predictive control (MPC), or more generally

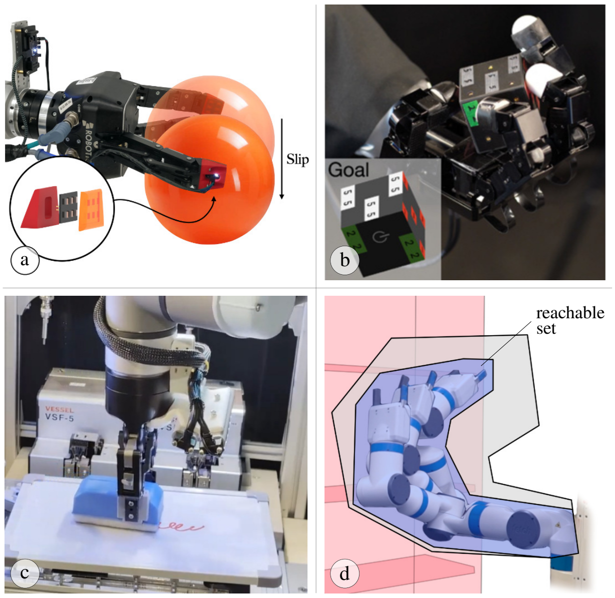

Figure 5. Examples of robust control mechanisms. (a) Reactive slip detection (Grover et al. 2021). (b) Predictive control for contact anticipation in dexterous in-hand reorientation (Suh et al. 2025) (© SAGE). (c) Compliance control for whiteboard cleaning (Kamijo et al. 2024) (© IEEE). (d) Control invariance and reachability analysis for robotic manipulation (Holmes et al. 2020). The blue area indicates the reachable set of the robot arm in a planning iteration.

receding-horizon control, optimize actions over a finite horizon and execute the first control input. For example, MPC-based hybrid force–motion control is employed to simultaneously regulate end-effector trajectories and contact forces during interaction, explicitly accounting for contact mode transitions within the prediction horizon (Jiang et al. 2025); a contact-implicit MPC is utilized that predicts motion and contact forces within a trust region, enabling stable local control under unilateral contact and friction constraints (Suh et al. 2025) (Fig. 5-b); tube-based MPC guarantees that the real trajectory remains within a bounded "tube" around the nominal one, ensuring performance under external disturbances and model mismatch (Nubert et al. 2020).

4.3.3 Compliance Compliance balances the control of position and force; higher compliance allows larger position deviations under external forces. It can be realized passively through mechanical design or actively within the control loop (Mason 2007); the former will be discussed in Section 4.5.1. Compliance control regulates how a robot responds to external forces during contact, allowing it to comply rather than rigidly enforcing a motion (Fig. 5-c). Impedance and admittance control represent the two classical approaches. Impedance control enforces compliance from motion to force by commanding motion trajectories that indirectly regulate contact forces (Hogan 1984),

$$M_d(\ddot{x} - \ddot{x}_d) + D_d(\dot{x} - \dot{x}_d) + K_d(x - x_d) = \tau_{\text{ext}}, \quad (7)$$

where $x \in \mathbb{R}^m$ denotes the task-space position (or pose) of the end-effector, x_d is the desired trajectory, and τ_{ext} is the external wrench arising from contact. The matrices M_d, D_d, and K_d are the desired inertia, damping, and stiffness parameters that define a virtual mass–spring–damper system. This equation specifies the closed-loop interaction dynamics: external forces induce bounded motion deviations governed by (M_d, D_d, K_d). On the other hand, admittance control does so from force to motion by computing the motion response to a commanded force (Dimeas and Aspragathos 2015).

Classical active compliance methods typically rely on accurate models or restrictive assumptions. More recently, model-free approaches have emerged, learning compliant behavior from human demonstrations or through reinforcement learning (Xu et al. 2026; Hou et al. 2025; Kamijo et al. 2024). For example, an adaptive compliance policy can adjust compliance both spatially and temporally from demonstrations for contact-rich tasks (Hou et al. 2025). Future directions may explore combining active compliance with passive mechanical compliance to enhance adaptability under contact uncertainty.

4.3.4 Invariance Invariance ensures that a system executes the behavior prescribed by a nominal controller while remaining within a certified safe set, such as bounds on force tracking error (Polverini et al. 2017) or admissible state regions. Like other control mechanisms, invariance-based methods do not resolve epistemic uncertainty in states or parameters. Rather, they achieve robustness by constraining system evolution so that bounded disturbances or modeling errors cannot drive the state outside a prescribed safe set. Formally, consider the closed-loop dynamics $x_{t+1} = f_\theta(x_t, u_t, w_t)$ under an output-feedback policy $u_t = \pi_t(h_t, \hat{\theta})$. A set $\mathcal{C} \subset \mathcal{X}$ is robustly forward invariant if

$$x_t \in \mathcal{C} \;\Rightarrow\; x_{t+1} \in \mathcal{C}, \quad \forall w_t \in \mathcal{W}.$$

Invariance-based control enforces this condition by modifying or projecting control inputs when the boundary of $\mathcal{C}$ is at risk of being violated, thereby guaranteeing constraint satisfaction despite aleatoric uncertainty and bounded modeling errors.

Practical approaches include reachability analysis, which conservatively overapproximates all possible evolutions to guarantee safety (Holmes et al. 2020) (Fig. 5-d), and control barrier functions (CBFs), which enforce invariance through real-time inequality constraints. When combined with control Lyapunov functions, CBF-based methods unify safety and goal achievement in an optimization framework, as demonstrated in tasks such as ball balancing (Wang et al. 2025). Despite their promise, invariance-based methods remain underexplored in robot manipulation. The main challenge lies in contact, which introduces discontinuities in the dynamics, arises spontaneously and unpredictably, and is notoriously difficult to model. It complicates the design of invariance constraints and corrective controllers. A promising direction is to employ data-driven approaches to approximate invariant sets or to learn barrier functions directly from demonstrations and interaction data.

4.4 Policy Learning

Parallel to the classical perception–planning–control pipeline, end-to-end policy learning has emerged as a model-free paradigm for manipulation. Learned policies can adapt to unstructured variations in the open world that are difficult to deal with analytically. Within the formalism of Section 3, these methods instantiate a parameterized policy π_ϕ and train it from data so that the robust performance functionals $\Gamma(\pi)$ remain high across an episodic distribution $\rho_N(\nu)$. In the rest of this section, we group policy-learning approaches according to four complementary ways (Fig. 6) in which they achieve robustness: *distribution-centric* methods shape the tasks, environments, and perturbations that π_ϕ is trained on (4.4.1); *architecture- and representation-centric* methods design policy classes whose inductive biases and temporal structure reduce sensitivity to nuisance variation and partial observability (4.4.2); *objective-centric* methods modify the learning objective to better align training with robust performance (4.4.3); and *adaptation-centric* methods allow policies to adjust at deployment through residual corrections, online updates, or test-time adaptation (4.4.4).

4.4.1 Distribution-centric robustness In the notation of Section 3, robustness is measured by functionals such as $\Gamma(\pi)$ evaluated over an episodic distribution $\rho_N(\nu)$, where $\nu = (\theta, x_0, G)$ encodes dynamics, initial conditions, and goals. Distribution-centric approaches act directly on this distribution: they choose how episodes are generated during training, while keeping the policy class Π and the learning objective fixed.

A first family of methods uses *domain randomization*. Instead of a single nominal environment, one defines a family of worlds θ (and sometimes x_0 and G) from a broad distribution. Policies trained under such variation can transfer to real manipulation with minimal tuning, as demonstrated for vision-based grasping appearance (Tobin et al. 2017) and for dexterous in-hand manipulation (Akkaya et al. 2019; Andrychowicz et al. 2020).

A second line achieves robustness via *data augmentation* on training trajectories. These methods keep the generator of ν fixed, but expand the empirical distribution $\hat{\rho}_N$ by transforming demonstrations according to task priors: prior work applies SE(2)/SE(3) transformations to scenes and actions to exploit invariance and equivariance (Florence et al. 2019; Mandlekar et al. 2023; Ameperosa et al. 2025), re-renders fixed trajectories from novel viewpoints to vary appearance (Zhou et al. 2023; Zhang et al. 2024), and retargets demonstrated trajectories to new object and scene configurations (Yu et al. 2023; Chen et al. 2023; Xue et al. 2025b). Beyond such structured priors, a related class injects calibrated noise into states and actions to mitigate covariate shift and compounding error, making the policy more robust (Laskey et al. 2017; Ke et al. 2021; Simchowitz et al. 2025).

A third class relies on *data scaling and task diversity*, as most visibly demonstrated in recent vision–language–action (VLA) models. Rather than explicitly parameterizing randomization or augmentations, these methods assemble very large, heterogeneous datasets of manipulation episodes spanning many robots, scenes, tasks, and language-specified goals, and train a single policy across this mixture (Brohan et al. 2022; Zitkovich et al. 2023; O'Neill et al. 2024; Kim et al. 2024). In the language of Section 3, robustness here is treated as generalization over a very broad empirical $\hat{\rho}_N$ that pools demonstrations and deployments across embodiments, tasks, and environments, with a single π_ϕ expected to maintain high $\Gamma(\pi_\phi)$ throughout this mixture.

4.4.2 Architecture- and representation-centric robustness A policy is a mapping $\pi : \mathcal{B} \to \mathcal{U}$ from beliefs to actions. Architecture- and representation-centric approaches pursue robustness by structuring this class: they constrain how information about states/observations (x_t, y_t) is encoded and how actions are parameterized, without changing the training distribution ρ_N or the learning objective. The goal is to make the mapping $b_t \mapsto u_t$ inherently less sensitive to nuisance variation, partial observability, and long horizons.

First, *perceptual and structural inductive biases* build a geometric state from raw observations. Keypoint-based representations encode objects and goals as a small set of 2D/3D landmarks and express tasks as geometric relations among these keypoints, so the policy operates in a low-dimensional space aligned with contacts and affordances rather than raw pixels (Manuelli et al. 2019; Qin et al. 2020; Huang et al. 2024). SE(3)-equivariant representations constrain internal features (and sometimes actions) to transform consistently under rigid motions of the workspace, so that rotating or translating the scene induces the same transformation in the encoded state and allows one controller to reuse a strategy across pose-shifted instances of a task (Simeonov et al. 2022; Ryu et al. 2022; Eisner et al. 2024; Wang et al. 2024). Graph-structured encoders treat objects, robot links, and goals as nodes with edges capturing spatial or relational constraints, and use message passing to learn local interaction rules that generalize to scenes with more objects, new goal configurations, and multi-object rearrangement (Li et al. 2020; Lin et al. 2022; Huang et al. 2022). Across these designs, robustness comes from aligning the learned representation with the geometry and relational structure of the task, so that pose- and relation-preserving changes in the scene tend to yield similar decisions and yields stable performance under these variations.

Second, *Temporal abstraction and hierarchy* represent actions as skills or options that span multiple timesteps. Extending the notation of Section 3, the policy acts in an extended action space Ω of options: a high-level policy selects a skill, and a low-level controller executes it until termination, so the decision-making problem becomes a semi-MDP with a shorter effective horizon (Sutton et al. 1998). Hierarchical RL methods learn both the skills and the high-level policy, often using subgoal states as the interface between levels and off-policy updates to remain sample efficient in continuous control (Nachum et al. 2018). In manipulation, skill-based approaches learn and plan with such temporally extended behaviors, such as predicting long-horizon discrete action sequences from a scene image for sequential physical reasoning (Driess et al. 2021), discovering reusable closed-loop skills from unsegmented demonstrations and training a meta-controller to compose them (Zhu et al. 2022), or chaining separate diffusion models for individual skills to solve unseen long-horizon tasks under constraints (Mishra et al. 2023). Robustness in this class arises from horizon reduction: the high-level policy makes skill-level decisions per episode, which reduces

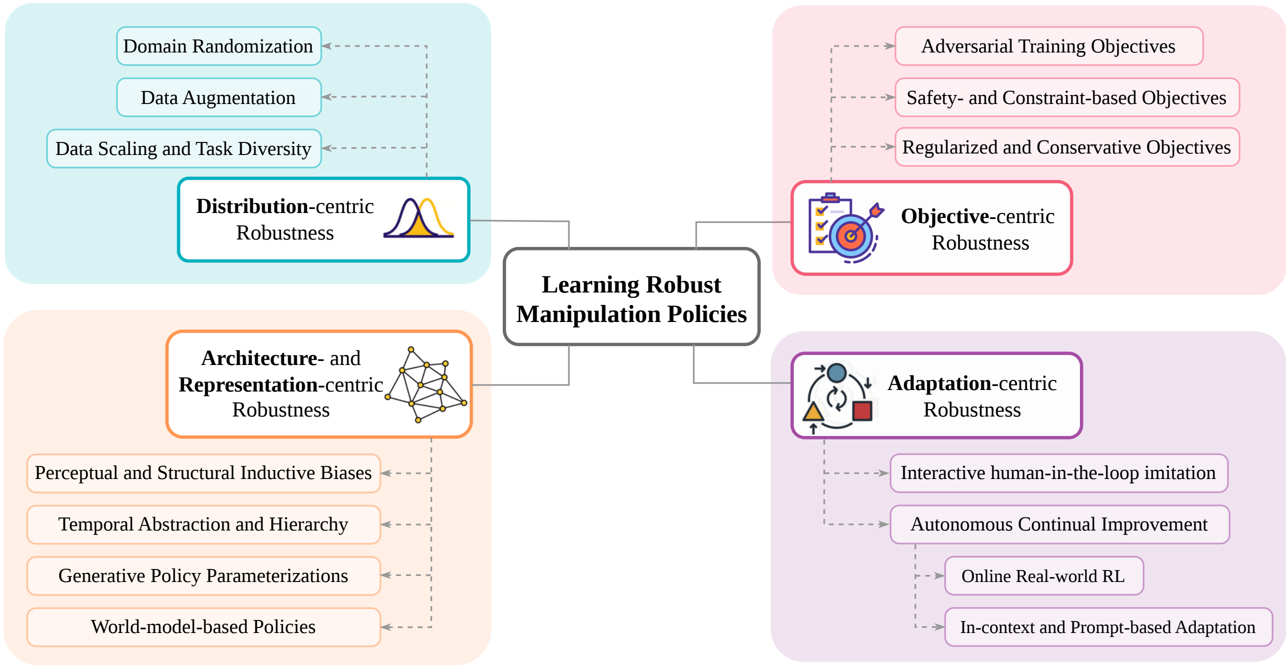

Figure 6. Taxonomy of mechanisms for learning robust manipulation policies.

compounding error and supports reuse of skills across task variations.

Generative policy parameterizations implement π_ϕ as a conditional generative process over short action sequences, most commonly using diffusion or flow-matching models for manipulation tasks (Chi et al. 2023; Black et al.). In the notation of Section 3, the policy still maps beliefs b_t to actions, but does so by starting from injected noise and iteratively refining a sample through a supervised denoising or flow integration process. Empirically, this iterative computation with noise injection induces an inductive bias that improves closed-loop robustness to covariate shift (Pan et al. 2025).

Lastly, *World-model-based policies* couple the controller with a learned dynamics model in a latent space. They maintain a latent state $z_t = e_\theta(o_{0:t})$ together with a predictive model $\hat{p}_\psi(z_{t+1} \mid z_t, u_t)$, and use short imagined rollouts in this latent space to reason about future states before committing to an action (Hafner et al. 2019; Ichter and Pavone 2019; Hansen et al. 2022; Hafner et al. 2023). In manipulation, such latent models are used for planning and runtime monitoring to score candidate actions under many predicted futures and reject those that would enter unsafe, constraint-violating, or high-risk regions (Liu et al. 2024; Nakamura et al. 2025; Sun and Song 2025).

4.4.3 Objective-centric robustness While distribution- and architecture-centric approaches shape what the policy sees and how it represents it, objective-centric robustness instead shapes what the policy optimizes for, embedding worst-case performance, safety, or conservatism directly into the learning objective.

Adversarial training objectives make robustness explicit by optimizing the min–max criteria of Eq. (6) rather than a nominal expected return. The policy is trained against an adversary that chooses disturbances, environment parameters, or competing behaviors to minimize task performance while the protagonist maximizes it. In practice, the adversary injects destabilizing forces or physical perturbations to grasps and object poses, so policies are trained on deliberate worst cases rather than randomly sampled disturbances (Pinto et al. 2017b,a; Jian et al. 2021).

Safety- and constraint-based objectives restrict which policies are admissible when we evaluate robustness. In the POMDP view of Section 3, one augments the control problem with constraint costs and searches for policies that achieve high task performance $\Gamma(\pi)$ while keeping these costs below a threshold, as in constrained MDP methods such as Constrained Policy Optimization (Achiam et al. 2017). Related work uses safety layers or control barrier functions that project or override the learned action when it would violate certified constraints (Dalal et al. 2018; Cheng et al. 2019).

Regularized and conservative objectives aim to improve robustness to epistemic uncertainty and limited data coverage by constraining how far the learned policy and value function extrapolate beyond the empirical episodic distribution $\hat{\rho}_N(\nu)$. Offline RL work frames this as "extrapolation error" and shows that robust performance typically requires (i) behavior regularization, which keeps π close to actions well supported by the data, and (ii) pessimistic value estimates that down-weight out-of-distribution actions (Levine et al. 2020; Kumar et al. 2020; Kostrikov et al. 2021; Fujimoto and Gu 2021). In real-robot manipulation, these ideas appear in conservative offline RL that improves over imitation on tabletop tasks using logs from safe operation (Zhou et al. 2022), in large-scale deployments that rely on heavily regularized off-policy updates for stable long-term waste-sorting with mobile manipulators (Herzog et al. 2023), and in regularized imitation methods that combine optimal-transport trajectory matching with a pull toward demonstration behavior to achieve few-shot real visual manipulation (Haldar et al.

2023). Across these examples, robustness comes from biasing learning toward regions of state–action space that are well covered and conservatively valued, making $\Gamma(\pi)$ less sensitive to model and dataset errors when evaluated over episodic variations.

4.4.4 Adaptation-centric robustness Adaptation-centric methods assume a base policy π_{base} has been trained under some episodic distribution $\rho_N^{\text{train}}(\nu)$, and focus on how its behaviour is modified using feedback from the deployed environment when the test-time distribution $\rho^{\text{test}}(\nu)$ differs from $\rho_N^{\text{train}}(\nu)$. Unlike distribution-centric approaches, they do not redesign ρ_N up front, but instead close the loop between deployment performance and further adaptation.

First, *interactive human-in-the-loop imitation* refines a pretrained policy at deployment using online expert feedback. In DAgger-style methods, imitation learning is reduced to online no-regret learning: at each iteration, the current policy is rolled out to generate states, the expert labels those visited states with preferred actions, these pairs are added to an aggregate dataset, and a new policy is trained on the union (Ross et al. 2011). This ensures that the learned stationary policy performs well under the state distribution it induces, rather than only on states seen in expert demonstrations, which directly addresses covariate shift in sequential decision making. Subsequent work adapts this idea to human experts in safety-critical or latency-limited settings (Kelly et al. 2019), to remote teleoperation for contact-rich manipulation (Mandlekar et al. 2020), to "on-the-job" deployment where autonomy and learning run continuously with humans stepping in on hard cases (Liu et al. 2022), and to kinesthetic delta corrections with force-aware residual policy via compliance (Xu et al. 2025).

Second, *autonomous continual improvement* adapts policies during deployment without relying on explicit expert labels. One branch uses online RL on the real robot: a warm-start policy (often from demonstrations or offline data) continues to collect rollouts on the hardware, and policy updates π_ϕ under the actual test-time episodic distribution, with mechanisms such as safety constraints, automatic or cheap resets, and simple reward specification to make this practical for contact-rich manipulation (Levine et al. 2016; Rajeswaran et al. 2017; Johannink et al. 2019; Xu et al. 2022; Luo et al. 2024, 2025). A second branch uses in-context and prompt-based adaptation: sequence models are trained on multi-task data so that, at test time, a short prompt of teleoperated demonstrations or recent trajectories for the current variation ν conditions the model, enabling few-shot adaptation to new objects, layouts, and goals without gradient updates (Duan et al. 2017; Valassakis et al. 2022; Fu et al. 2024).

4.5 Mechanical Intelligence

Manipulation robustness can also be achieved through the robot's physical embodiment ("body"), complementing the contributions of the "brain", sensors and actuators discussed above. This idea is grounded in the concept of *morphological computation*, where physical structures perform functions that would otherwise require explicit control (Paul 2006). The body (materials, geometry, actuation) can shape contact dynamics P_θ so that a wider set of states X_t or actions U_t leads to success, and so that errors in perception, modeling, and execution have smaller consequences. More specifically, mechanical intelligence often improves robustness by designing the environment/robot parameter θ (e.g., joint compliance, stiffness, contact properties), which directly changes P_θ. The same controller or policy thereby achieves higher expected or worst-case performance $\Gamma(\pi)$ under uncertainty and variation. In the following, we outline complementary principles of mechanical intelligence, each corresponding to a different way of shaping the system dynamics P_θ: *passive compliance* shapes local impedance at the contact interface and dissipates disturbances (Section 4.5.1); *adhesion* modifies contact properties and the feasible wrench set (4.5.2); *postural synergy* constrains the reachable configuration or action space to low-dimensional subspaces (4.5.3); and *morphology adaptation* alters global structure or operation modes across episodes/tasks to keep objectives tractable under changing conditions (4.5.4).

4.5.1 Passive Compliance Passive compliance enhances manipulation robustness by physically reducing the sensitivity to disturbances and contact uncertainty through compliant, soft surface material, before they propagate through the control loop. It can be viewed as a mechanical counterpart to active compliance (Section 4.3.3) that shapes the interaction dynamics prior to control. Specifically, compliant embodiments reduce the sensitivity of the transition dynamics P_θ to disturbances (W_t, V_t) and contact modeling errors by allowing bounded motion in response to external forces, in contrast to active compliance, which relies on feedback control. As a result, safe and adaptive interaction can be achieved with reduced dependence on precise sensing, modeling, and high-bandwidth control. This principle is widely observed in nature and robotics. Dolphins, for example, use marine sponges as protective tools while foraging on rocky seabeds (Fig. 7-a), filtering unpredictable impacts caused by contact with scattered rocks (Mann et al. 2008). In robotic manipulation, passive compliance is commonly used to tolerate misalignment and reduce impact forces in assembly tasks (Drake 1978), as well as in soft grippers based on compliant materials or tendon-driven structures that conform to object geometry under uncertainty (Deimel and Brock 2013). Hybrid designs further combine compliance with controllable stiffening, such as granular jamming grippers. They enable compliant contact exploration followed by mechanically conforming to and securing the object (Shintake et al. 2018; Amend et al. 2012).

4.5.2 Adhesion Adhesion enhances manipulation robustness by modifying the robot-object contact surface to sustain strong tangential (shear) forces with minimal normal force, which is particularly suitable for delicate, thin, or flat objects. Adhesion effectively changes θ, especially the contact properties of the end-effector surface. It alters the contact dynamics P_θ and enables patch-based attachment modalities beyond point-contact, friction-limited force closure. Common mechanisms include gecko-inspired dry adhesion and vacuum-based suction (Shintake et al. 2018). Gecko-inspired adhesives exploit van der Waals forces to generate strong shear interactions across a wide range of surface textures without requiring active control (Fig. 7-b);

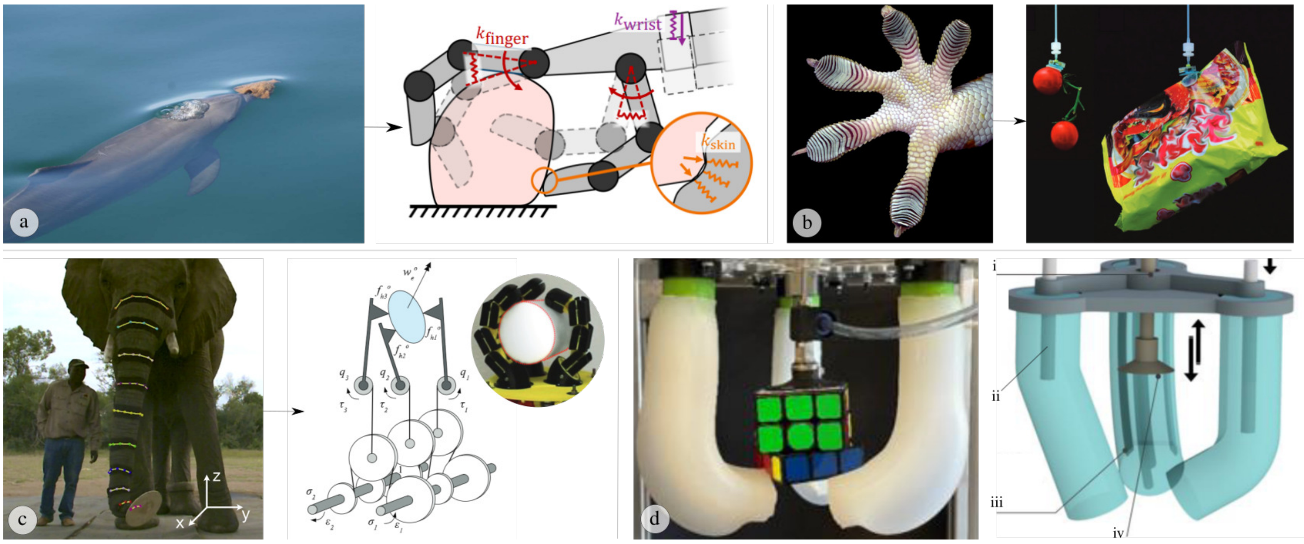

Figure 7. Examples of hardware mechanisms for robust manipulation. Hardware intelligence in robotics is often inspired by biological strategies. (a) Passive compliance: Dolphins wrap marine sponges around their beaks for protection (Mann et al. 2008) (© PloS one), while robotic hands employ compliance in finger joints, wrists and skin (Junge and Hughes 2025) (© Springer Nature). (b) Adhesion: gecko feet (image courtesy of Prof. Kellar Autumn, Lewis and Clark College) inspire adhesive grippers (Song et al. 2017) (© National Academy of Sciences). (c) Postural synergy: elephants control their trucks with muscular synergies (Dagenais et al. 2021) (© Elsevier), a principle adopted in adaptive robotic hands (Catalano et al. 2014) (© SAGE). (d) Morphology adaptation: a robotic gripper with reconfigurable fingers, adjusted by rigid rods (ii), adapts its geometry to target objects (Pagoli et al. 2021) (© IEEE).

such adhesives have been used to grasp bulky objects in microgravity (Jiang et al. 2017), demonstrating robustness to object misalignment. Vacuum-based suction is another widely adopted strategy utilizing adhesion. Many suction cups also incorporate bellows that provide passive compliance, allowing the system to accommodate pose errors and conform to local surface curvature, although performance degrades on porous, dusty, or highly irregular surfaces. To mitigate these limitations, hybrid systems combine multiple grasping modalities, such as multi-affordance manipulators integrating suction with a parallel-jaw gripper (Zeng et al. 2022); this mechanical redundancy improves robustness and extends applicability across diverse environments.

4.5.3 Postural Synergy Beyond shaping the contact interface through impedance (passive compliance) or contact modalities (adhesion), robustness can also be improved by encoding coordination within the embodiment itself via mechanically enforced postural synergies. Biological systems coordinate many degrees of freedom into low-dimensional units known as synergies: rather than actuating each joint or muscle independently, the nervous system activates groups of elements as functional units, simplifying control while preserving adaptability. For example, an elephant trunk comprise over 40,000 muscles yet reduces biomechanical complexity through motion primitives and muscular synergies (Dagenais et al. 2021) (Fig. 7-c). Postural synergy can be interpreted as a structured restriction of the control and state spaces, e.g., by constructing low-dimensional action spaces and concentrating reachable configurations on a task-relevant manifold. This reduces the control complexity, and therefore the sensitivity of the closed-loop transition dynamics to disturbances and contact-model mismatch. Inspired by this principle, robotic hands use a single actuator to drive multiple joints through underactuated tendons and elastic couplings, allowing the hand to passively conform to object shapes and achieve reliable grasps across diverse geometries with reduced sensing and planning demands (Catalano et al. 2014). Complementarily, low-dimensional hand posture subspaces have been constructed to facilitate planning and control of complex robotic hands (Ciocarlie and Allen 2009).

4.5.4 Morphology Adaptation Robotic systems can adapt their embodiment by morphing geometry, stiffness distribution, or kinematic structure across diverse environments and tasks (Liang et al. 2026). Complementary to passive compliance and adhesion, which shape the contact interface locally, and postural synergy, which forms coordination within an embodiment, morphology adaptation reshapes the embodiment at a global level beyond adapting the robot parameterization θ. This effectively selects a morphology that makes the same high-level objective more tractable under changing episodes. In soft robotic hands, for example, the bending location and range can be adjusted by inserting a stiff rod into the center of each finger (Pagoli et al. 2021) (Fig. 7-d). This flexibility enables the end-effector to switch between grasping modes and maintain robust performance across object types without complex sensing or control. PolyBot (Yim et al. 2000) further illustrates modular reconfiguration: detachable modules autonomously rearrange to accommodate new workspace constraints, supporting fault tolerance and effective manipulation in unstructured environments. At smaller scales, reconfigurable microrobot swarms (Xie et al. 2019) can transition between collective modes (e.g., liquid, chain, vortex) by tuning external magnetic fields, allowing the system to adapt its interaction mode to fluidic disturbances while preserving task intent.

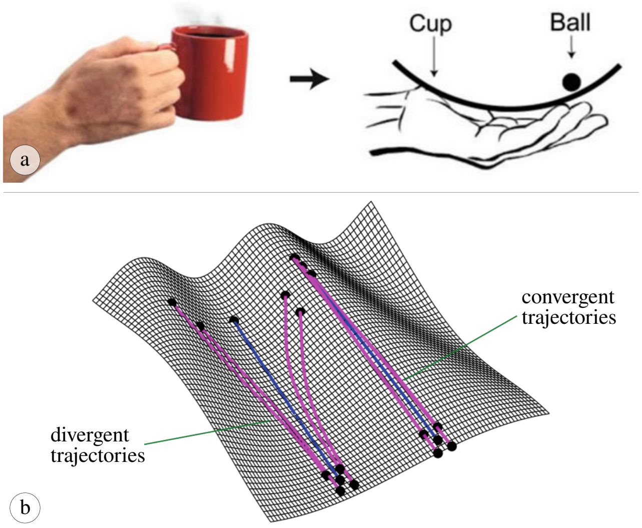

Figure 8. Examples of robustness evaluation methods. (a) Margin to failure: coffee inside a mug has a safety margin below the rim of the mug (Sternad and Hasson 2016) (© Springer). (b) Convergence metrics: self-correcting behaviors can be exploited in the system dynamics using convergence metrics (Kong and Johnson 2019) (© Springer).

5 Evaluation Methods of Manipulation Robustness

Evaluating manipulation robustness requires quantitative criteria that reflect a system's capability to achieve its task objectives under challenges. In this section, we first present canonical evaluation methods based on the probability of task success under epistemic uncertainty, aleatoric uncertainty, and episodic variation. We then introduce several proxy measures that instantiate or approximate this canonical notion of robustness in specific manipulation scenarios.

Given a manipulation task specification together with domains of epistemic uncertainty, aleatoric uncertainty, and episodic variation, the most widely used robustness evaluation protocol is to measure empirical task success or failure (Mason 2018). This approach, commonly referred to as *Monte Carlo robustness tests*, executes the system repeatedly under sampled model mismatch $(\hat{\theta}, \theta)$, stochastic disturbances w_t and v_t, and episodic variations ν, and estimates the probability of achieving the task goal. Concretely, robustness is the expected probability of goal satisfaction under the challenges,

$$\mathbb{E}_{\nu \in \mathcal{N},\, w_t \in \mathcal{W},\, v_t \in \mathcal{V}} \left[\mathbf{1}\{x_T \in G\} \mid (\hat{\theta}, \theta) \right], \tag{8}$$

often approximated by empirical success rates $n_{\text{success}}/n_{\text{total}}$ computed over multiple trials. This evaluation directly aligns with the robustness formulations introduced in Section 3.

Following this canonical protocol, example evaluations include success rates in object pose estimation under perception noise and occlusion (Mandlekar et al. 2023); success rates or expected returns in reinforcement learning, which often reduce to goal satisfaction indicators, evaluated across episodic variations such as different initial object placements or object instances (Luo et al. 2025; Andrychowicz et al. 2020); and success under adversarial perturbations, including external disturbances applied to manipulated objects or injected joint-level perturbations during execution (Jian et al. 2021). Variants of this approach include systematic parameter sweeps, where parameters in θ (e.g., inference latency, friction coefficients, lighting conditions) are varied across controlled ranges and performance is plotted as a function of the parameter (Chi et al. 2023).

While success probability under challenges provides a canonical, task-level notion of robustness, it is often insufficiently fine-grained for answering more specific questions about *how* a manipulation system withstands challenges. In many scenarios, one is not only interested in whether a task succeeds, but also in the structure and degree of tolerance to disturbances, uncertainty, or variation. Accordingly, we introduce several representative proxy evaluations in the remainder of this section. The proxy evaluation methods provide structured and interpretable approximations of the canonical success probability in Eq. (8), tailored to specific manipulation mechanisms and failure modes. Depending on the application, they may be used either as surrogate objectives integrated into planning, control, or learning for robust manipulation synthesis, or as diagnostic tools for robustness analysis.

5.0.1 Closure Properties Closure properties characterize how contact forces and geometric constraints prevent an object from deviating from a desired configuration or set of configurations. These properties are typically analyzed from two different perspectives: *local* conditions at the contact and *global* conditions within the object's configuration space.

Local metrics derived from force, form, or object closure quantify the instantaneous robustness of quasistatic manipulation via wrench-space or geometric analysis. These metrics can be evaluated either analytically or empirically. Analytical approaches, commonly referred to as grasp quality metrics (Ferrari et al. 1992), compute robustness using physics-based models to determine the system's ability to resist external disturbances. In contrast, empirical approaches approximate these analytic metrics by learning from data. For instance, grasp datasets labeled with analytic quality values are used to train neural networks that predict robustness directly from visual or sensory inputs (Mahler et al. 2017), enabling scalable estimation in unstructured environments.

While local metrics assess stability via instantaneous force or geometric conditions, many complex manipulation tasks depend on the global characteristics of the configuration and contact spaces—specifically, how reachable, connected, or "trapped" an object remains within its environment. Topological analysis captures such global invariants, providing a perspective complementary to purely local evaluations. For example, energy-bounded caging (Mahler et al. 2018) defines robustness in terms of the size or persistence of the configuration-space region that keeps the object contained. In this context, a larger connected component implies a greater structural tolerance to uncertainty or disturbances.

5.0.2 Margin to Failure A related evaluation to the closure methods is the margin to failure, which quantifies robustness by measuring the distance between a system's operating state and the boundary of task failure. Intuitively, systems maintaining larger margins are more robust, as they can tolerate greater disturbances, modeling errors, or stochastic noise before a failure occurs. Consider the task of placing a cup of coffee on a coaster (Fig. 8-a): precise positioning is secondary as long as the liquid level remains safely below the rim. If the goal region G is defined as the set of states where all liquid remains inside the cup, its complement G^c corresponds to failure (spillage). While a binary reward only detects the occurrence of failure, a safety margin quantifies the resilience of the current state:

$$\sigma_{\text{margin}} = \text{dist}(x, \partial G), \tag{9}$$

where $\text{dist}(\cdot)$ denotes a task-appropriate distance metric and ∂G is the boundary of the goal region. Positive values indicate the system is within G, with larger values implying greater robustness.

In neurophysiology, this distance is often defined through energy, i.e., the energy required for the system to transition from a safe state x to failure G^c (Hasson et al. 2012). Humans intuitively maintain such margins by relying on simplified internal models rather than exact dynamics, selecting control strategies that preserve safety margins (Sternad and Hasson 2016). Similar concepts have been applied to evaluate robotic manipulation robustness (Dong et al. 2024). Analogous ideas appear in force- and wrench-based robustness metrics, such as grasp stability measures that quantify the maximum external wrench a grasp can resist before slippage (Roa and Suárez 2015). Extending the concept over time leads to the notion of a "safety tube", a region surrounding a nominal trajectory within which the system can remain despite bounded disturbances (Fox et al. 2006; Nubert et al. 2020).

5.0.3 Convergence and Divergence Metrics In dynamically complex manipulation tasks such as tossing, catching, or balancing, robustness often arises from the intrinsic structure of the system dynamics rather than from geometric constraints or explicit safety margins. Humans routinely exploit such dynamics by inducing self-correcting behaviors in which small perturbations are naturally compensated, allowing motion to remain stable under bounded aleatoric disturbances $w \in \mathcal{W}$. From an evaluation perspective, this form of robustness can be captured by convergence or divergence metrics (Bazzi and Sternad 2020), which quantify how trajectories evolve under perturbations (Fig. 8-b). For example, the largest eigenvalue of the symmetric part of the system Jacobian measures the local rate of trajectory divergence or contraction, with negative values indicating stabilizing, disturbance-rejecting dynamics. Such metrics can be incorporated into constrained optimization or planning formulations, e.g., as constraints on contact dynamics, to favor actions that induce convergent behavior and naturally funnel executions back toward desired outcomes.

6 Discussions

In this section, we synthesize key insights that emerge from the literature reviewed in this paper. We focus on cross-cutting principles underlying manipulation robustness, strategies for managing failure, and broader perspectives and takeaways for practitioners.

6.0.1 Principles of robustness mechanisms Across the diverse mechanisms surveyed in this paper, several recurring principles emerge. These principles cover perception, planning, control, learning, and hardware, and together enrich the toolbox available to practitioners developing robust manipulation systems.

One of such principles is *compliance*, realized both actively through control (Section 4.3.3) and passively through hardware and materials (4.5.1). Compliance mitigates sensitivity to contact uncertainty by absorbing disturbances, and enables safe interaction under modeling error. Another recurring principle is *invariance*, where behaviors or representations are designed to remain unaffected by uncertainties or variations, as seen in closure-based grasping (4.2.4), invariant perception features (4.1.4), and control barrier functions (4.3.4). *Mechanics*-driven strategies achieve robustness by shaping interactions to exploit task and environmental structure. Motion strategies, funnels, and extrinsic dexterity (4.2.3) deliberately leverage gravity, contact, and geometry rather than tackling the challenges through precise control.

The mechanisms reviewed here are not exhaustive. Biological systems offer additional inspiration (Kitano 2004), frequently leveraging *modularity* to localize disturbances and prevent their propagation. For example, ants dynamically reassign roles during cooperative transport (Gelblum et al. 2015), a strategy that has inspired modular robotic systems with improved adaptability and fault tolerance (Yim et al. 2000; Xie et al. 2019). Another principle is *redundancy*, which provides multiple pathways (often more than needed) to task success and allows temporary failure or degradation in one component to be compensated by others. Examples discussed in Section 4 include multi-sensor fusion (Li et al. 2023), dual end-effectors (Zeng et al. 2022), and alternative task-level plans (Siméon et al. 2004). Together, these principles emphasize that robustness rarely arises from a single mechanism, but from the coordinated interaction of many.

At a higher level of abstraction, the robustness mechanisms surveyed in this paper can be broadly grouped into two complementary strategies for dealing with uncertainty and variation: *reduction* and *toleration*. Reduction primarily targets epistemic uncertainty by improving the robot's internal model or state estimates, as exemplified by active and Bayesian perception (Section 4.1.1, 4.1.3), reasoning over uncertainties in planning (4.2.1), and adaptation-centric learning that updates policies using deployment data (4.4.4). In contrast, toleration accepts that aleatoric uncertainty and episodic variation cannot be eliminated and instead seeks to make task execution insensitive to them, through mechanisms such as active and passive compliance (4.3.3, 4.5.1), motion strategies and closure in planning (4.2.3, 4.2.4), or invariant perceptual representations and controllers (4.1.4, 4.3.4). Robust manipulation systems typically integrate both strategies, reducing uncertainty where possible while deliberately designing interactions, embodiments, and policies that remain effective when irreducible uncertainty inevitably persists.

6.0.2 Manipulation failure management Failure management is a central aspect of manipulation robustness. Broadly speaking, two complementary strategies can be distinguished: *failure prevention* and *failure recovery*. The relative importance of each depends strongly on task semantics. In tasks where temporary failure is acceptable, such as regrasping rigid objects after a drop, recovery mechanisms may suffice. In contrast, tasks involving fragile objects or irreversible consequences often demand predictive failure avoidance, as even brief failure may be unacceptable.

Failure prevention aims to avoid entering failure states altogether. Examples include down-weighting risky rollouts in receding-horizon control (Section 4.3.2), synthesizing robust grasps using closure-based analysis (Howard and Kumar 1996; Bicchi 1995; Pereira et al. 2004), or exploiting compliance through control and hardware design (4.3.3, 4.5.1). These approaches emphasize anticipation and constraint satisfaction. By contrast, failure recovery focuses on restoring task execution once failure has occurred. Model-based approaches can stabilize payloads after actuator failure (Michael et al. 2011), while recent learning-based systems demonstrate the ability to acquire recovery behaviors from imperfect or failed demonstrations (Barreiros et al. 2025). Robust manipulation systems often integrate both strategies, balancing prevention with recovery to handle a wide range of contingencies.

6.0.3 Robustness and Performance Trade-offs Manipulation performance is inherently context-dependent. A system may exhibit high performance in controlled laboratory settings, yet experience sharp degradation in unstructured real-world environments. Systems optimized for narrow conditions can achieve exceptional speed or precision, but often do so at the expense of robustness when those conditions change. In nature, star-nosed moles exemplify this trade-off: their specialized nasal appendages enable rapid prey detection in wetland tunnels, but such specialization may be fragile under environmental shifts (Catania and Kaas 1996). By contrast, some systems with moderate peak performance may trade efficiency or accuracy for robustness. Compliant robotic grippers, for example, adapt well to variations in object shape or mass, but typically sacrifice positional precision (Deimel and Brock 2013). These observations suggest that progress in manipulation should be evaluated not only by peak performance under ideal conditions, but by the ability to sustain reliable function under imperfect and unpredictable in-the-wild challenges.

6.0.4 Insights from beyond Manipulation Robustness Robustness research in domains beyond manipulation offers valuable insights. As mentioned in Section 2, locomotion robustness is particularly relevant, as locomotion can be viewed as a form of "self-manipulation" (Johnson et al. 2016; Mason 2018). Techniques such as adversarial training for controller robustness (Shi et al. 2024), widely explored in locomotion, remain comparatively underutilized in manipulation and represent a promising direction. Similarly, biological inspirations such as gecko-inspired adhesives, originally studied in climbing and locomotion, have motivated gripper designs (Jiang et al. 2017). Despite these overlaps, manipulation presents unique challenges: unlike locomotion, which often reduces interaction to discrete foot–ground contacts, manipulation involves rich, multi-body interactions among the hand, object, and environment, with diverse contact types and combinatorial complexity that significantly complicate robustness analysis and design.

6.0.5 Beneficial Role of Perturbations Finally, perturbations, typically viewed as detrimental, can in some cases enhance robustness. In granular media, small vibrations can break clogging arches when pouring sand into a funnel (To et al. 2001). The vibrating bowl feeder discussed in Section 4.2.3 provides another example, where controlled vibrations guide parts toward consistent configurations (Fig. 4-c3). Analogously in manipulation, slight, intentional perturbations can dislodge unstable contacts, reduce excessive contact forces, or guide objects out of shallow local minima. Everyday actions such as wiggling a Jenga block or flicking an omelette exemplify how controlled perturbations stabilize interactions rather than destabilize them. These observations highlight that robustness is not always achieved by tackling uncertainties or variations, but sometimes by strategically exploiting them.

7 Challenges and Open Problems

The frameworks and principles presented thus far provide a basis for understanding how robustness arises, yet they also make clear that essential scientific and engineering questions remain unanswered. This section identifies some of the open problems and discusses the challenges that must be overcome to achieve manipulation robustness under real-world challenges.

7.0.1 Benchmarking and Evaluation of Manipulation Robustness Effective comparison of manipulation robustness across different methods requires well-designed benchmarks and consistent evaluation methods. Simulation-based benchmarking (Tao et al. 2024; James et al. 2020; Zhu et al. 2020) allows experiments under identical conditions and facilitates large-scale testing, yet faces the persistent challenge of the sim-to-real gap that limits physical realism and transferability, as well as being confined to a very narrow distribution of tasks and environments. Real-world benchmarking efforts, including standardized object sets (Calli et al. 2017), manipulation competitions (Correll et al. 2016; Kasper et al. 2012), cloud-based robotic platforms (Zahid and Pokorny 2024), and community-run distributed infrastructures (Chen et al. 2026; Atreya et al. 2025), aim to address this gap, yet each makes trade-offs among task diversity, scalability, accessibility, authenticity, or consistency. Future progress depends on balancing these aspects through advances in computation, communication, and standardized cross-community protocols. Most importantly, the existing simulation or real-world benchmarks rarely treat robustness as an explicit evaluation criterion, instead evaluating performance primarily under relatively fixed conditions. Fair comparison further requires robustness evaluation methods that quantify system performance. While analytical and empirical methods have been developed for specific manipulation tasks (Section 5), a unified framework that systematically captures real-world uncertainty or variation remains an open challenge.

7.0.2 Analytical and Empirical Robustness Analytical robustness methods reflect Plato's philosophy of abstraction, seeking general principles and guarantees through formal modeling and analysis; empirical methods follow Aristotle's practical spirit, emphasizing performance validated through data and experience. The field continues to debate whether progress in robotics lies in one paradigm or their synthesis (Amato et al. 2025), and robustness as a core robotic topic follows the same divide. Data offer robots experiential knowledge that models alone cannot provide, enabling robustness to emerge from interaction, much as humans develop know-how before know-why through sensorimotor learning. Yet unlike human toddlers, robots are not bound to start from scratch. They can inherit models and priors designed by humans, effectively knowing why before knowing how. Such models allow robots to adapt behavior in a data-efficient way while mitigating the limitations of purely data-driven methods, which are often sensitive to aleatoric uncertainty and distribution shift. Consider the game of Jenga: humans learn to play robustly not only through trial and error but also by leveraging internal models of perception, planning, and control. Similarly, the future of manipulation robustness likely lies in unifying data-driven learning and model-based reasoning—integrating know-how and know-why into robot systems.

7.0.3 Towards Human-Level Manipulation Robustness

Achieving human-level manipulation robustness remains a grand challenge in robotics. Evidence suggests that such robustness emerges from the integration of multiple complementary strategies, such as extensive lifetime experience, accurate world models grounded in physical reasoning, domain-specific embodiment, and rapid adaptation. For example, crows bend hooks to retrieve food, effectively co-designing hardware and control in the wild (Hunt and Gray 2004); such behavior cannot be explained by mechanics or control alone. Humans similarly integrate rich lifetime priors with online exploration and adaptation, inspiring approaches such as reinforcement learning bootstrapped with prior data (Luo et al. 2025). Yet, synthesizing all these strategies in a single robotic system remains rare and challenging. Other principles evident in animals and humans, such as redundancy in planning and sensing, are gaining attention as essential components for manipulation robustness. Reaching animal-level manipulation robustness already exceeds current robotic capabilities, but is prospective; attaining human-level robustness represents a far greater leap and the ultimate aspiration for robotic manipulation.

8 Conclusion

This review makes advances towards a systematic understanding of manipulation robustness by defining the concept, formulating the problem, and revisiting prior work across subfields. We surveyed representative mechanisms and evaluation methods, highlighting the core principles that enable robustness in robotic manipulation and providing insights for future practitioners interested in this topic. Despite decades of progress, achieving biological-level manipulation robustness on robotic systems under real-world uncertainties and variations remains a central challenge. Bridging this gap will require not only improved algorithms and hardware, but also clearer formulations of robustness and more consistent evaluation methods. We hope that this survey will help organize future research efforts toward robotic manipulation systems that approach the robustness routinely demonstrated by humans and animals in the physical world.

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